21 Whiting in Skagerrak and Kattegat

whg.27.3a – *Merlangius merlangus* in Division 3.a

21.1 General

21.1.1 Stock definition

There is a paucity of information on the population structure of whiting in Division 3.a (the Skagerrak-Kattegat area). No genetic or otolith-based surveys have been conducted. Tagging of whiting has previously been undertaken, but these data need to be re-examined. Results from previously modelled survey data (SURBAR) were inconclusive regarding independent population dynamics in Division 3.a in comparison with the North Sea, presumably due to the need of age readings in 3.a (age information used in SURBAR was borrowed from Subarea 4). The drop in landings in the beginning of the 1990s gives, however, an indication of local stock structure as this reduction was not paralleled by any similar event in the North Sea (Table 21.1). There are also findings of locally spawned whiting eggs in Kattegat 3.a (Börjesson *et al.*, 2013).

21.1.2 Ecosystem aspect

A summary of available information on ecosystem aspects is presented in the Stock Annex last updated at ICES WKDEM (ICES, 2020).

21.2 Fisheries data

21.2.1 Catches

Whiting landings in Division 3.a have declined in recent decades from over 20 000 tonnes in the 1980s to 352 tonnes in 2024 (Table 21.1). Denmark is catching most of the whiting in the area; Sweden and Norway follow with considerably less amounts. The Danish industrial fleet (main target species: sprat) is landing 40–80% of whiting in the area, but this percentage has dropped to 13% in 2023 and 16% in 2024. Information on whiting 3.a is uploaded to InterCatch by Sweden, Denmark, Norway, Germany, and the Netherlands. Discard estimates are available since 2002. A summary of available information on fisheries and information on derivation of discards is presented in the Stock Annex (last updated during the WKDEM 2020 benchmark (ICES, 2020).

21.2.2 InterCatch data

The estimation of discards is done using InterCatch data. In 2023 and 2024, the ICES estimated catch was equal to 726 and 854 tonnes, respectively. They correspond to landings and discards (imported or raised) as shown in Figure 21.1 and 21.2 and Table 21.2. The raising of discards for unsampled strata was done, assuming a discard rate equal to a weighted mean of reported discard rates, with weights equal to the total landings in tonnes. The raising is done by grouping all fleets by area. The industrial fleet, responsible for a substantial part of the landings (13% in 2023 and 16% in 2024), does not have any discards. The landings and estimated discards are shown in Figure 21.3 and Table 21.1.

21.3 Survey index

A combined survey index was derived using four bottom trawl surveys that operate in the area, namely the two international bottom trawl surveys (NS-IBTS (Q1 and Q3) and BITS (Q1 and Q4)) and two Danish national bottom trawl surveys targeting cod and sole both conducted in Q4 (Figure 21.4).

The survey index calculation is described in the stock annex, here a short description is given. Predictions of a Tweedie Generalised Additive model on a fine grid are used to estimate the biomass index. The model is described by the following equation:

$$\log(\mu i) = \text{Gear}(i) + f_1(\text{lon}_i, \text{lat}_i) + f_2(\text{timeOfYear}_i, \text{lon}_i, \text{lat}_i) + f_3(\text{time}_i, \text{lon}_i, \text{lat}_i) + f_4(\text{depth}_i) + U(i)_{\text{ship:gear}} + \log(\text{HaulDur}_i),$$

which includes a time-invariant spatial effect (f_1), a seasonal repeating pattern (f_2), a space-time interaction effect (f_3) that can capture smooth changes over longer time scales, a smooth function of depth (f_4), a fixed gear effect and random effects for the interaction between ship and gear. Finally, the model includes an offset term of the logarithm of haul duration that corresponds to the assumption that catch is proportional to haul duration.

The prediction of the biomass index in Quarter 1 including survey information up to 2023 is used for advice purposes in 2025 and 2026. The prediction of the biomass index in Quarter 1 using survey data up to 2024 is shown in Figure 21.5 and Table 21.3. Quarter 1 is used as it covers the early part of the spawning period of whiting in the Greater North Sea (González-Irusta and Wright, 2017). Maps of survey biomass show a slightly higher abundance in the Skagerrak area compared to 2023 and an increase in the index in the most recent years. The retrospective analysis, calculated by leaving out the last 1–5 years of available data, is shown in Figure 21.6. The Leave-one-survey-out analysis of the standardised Quarter 1 biomass index is shown in Figure 21.7.

21.4 Analysis of stock trends/assessment and advice

21.4.1 Exploratory assessments

In 2016, an exploratory SURBAR analysis has been performed that showed that internal consistency was virtually absent, impeding cohort analysis for the stock (ICES, 2016). The main conclusion from the SURBAR analysis was that the lack of internal consistency in the available survey indices (Figure 12.1.6 in ICES 2016) prevents an analytical assessment. This inconsistency could be related to a) age reading problems, and/or b) a mixture of several stock components leading to unaccounted migrations.

During the WKDEM 2020 benchmark (ICES, 2020) there was an attempt to do an assessment using the surplus production model in continuous time (SPiCT). The estimated uncertainty was very high and none of the scenarios could be used to provide advice for the stock. During WGNSSK 2022 and in preparation of WGNSSK 2025, an exploratory SPiCT assessment had similar issues and was not put forward to provide catch advice.

In preparation of WGNSSK 2025, a transition from the *rb*-rule to the *rfb*-rule was investigated, addressing the lack of length information from the industrial fleet by utilizing length distributions from survey data. A document detailing this approach is in preparation for external review in 2025.

21.4.2 Advice

During the last benchmark of whiting in Division 3.a. in 2020 (ICES WKDEM, 2020), the stock was raised from category 5 to category 3 (ICES, 2018). In 2020, the advice was based on the trends of a new combined survey index, introduced in the benchmark, using the "2-over-3 rule". In 2022, ICES moved away from the "2-over-3" rule as it was shown to be not precautionary in large scale Management Strategy Evaluations. From 2022, a set of empirical rules are used instead, summarised in the WKLIFE-X report (ICES, 2021).

Whiting in Division 3.a lacks representative length distributions from the catch, as very few samples from the industrial fishery are collected. The industrial fleet is responsible for a considerable part of the landings and uses gear with small mesh sizes, thus, catching smaller whiting compared to the other fleets. The consequence is that the only applicable empirical rule is the "*rb-rule*". This consists of a multiplier "*r*" that captures the survey index trends, a biomass safeguard *b* that is below 1 if the stock biomass index is lower than a trigger reference point $I_{trigger}$, defined as 1.4 I_{loss}, where I_{loss} is the lowest observed index over the index time series up to the benchmark. Finally, a multiplier $m = 0.5$ is used to ensure that the rule is precautionary. The calculations are summarised in Table 21.4

The advice for whiting in Division 3.a is biennial and no updates have been made this year. For whiting in Division 3.a, ICES advises that when the precautionary approach is applied, catches in each of the years 2025 and 2026 should be no more than 455 tonnes.

The history of ICES advice, advice basis and agreed TAC is shown in Table 21.5.

21.5 Issues for future benchmarks

21.5.1 Data

In the routine surveys, IBTS quarter 1 and quarter 3 in Division 3.a, biological data are collected for this species, in particular otoliths for aging and maturation information. These can be used to understand growth and maturity patterns of the population in this area.

A genetic study is underway to assess the population structure of whiting in Division 3.a using samples from both the North Sea and the Kattegat/Skagerrak.

21.5.2 Assessment

There have been several attempts to assess the stock using the surplus production model in continuous time (SPiCT). Two issues make such an assessment difficult: 1) the whiting 3.a biomass index is based on a single survey model that combines multiple surveys from different quarters of the year. As a result, it was recommended not to include year as a factor in the model, as doing so would introduce artificial discontinuities between Q4 of year x and Q1 of year x+1. By excluding year as a factor, the resulting index series is smoothed over time. This smoothing introduces temporal autocorrelation in the index, which the current SPiCT model is not designed to handle and that may cause issues when fitting SPiCT(ICES, 2020); 2) the drop in landings in the beginning of the 1990s is uncorrelated to the index time-series, which create large uncertainty in the estimate of stock status in SPiCT.

In 2024, a transition from the *rb*-rule to the *rfb*-rule was investigated, addressing the lack of length information from the industrial fleet by utilizing length distributions from survey data. A document detailing this approach is in preparation for external review in 2025.

21.5.3 Forecast

No forecast is made for the stock.

21.6 References

Börjesson, P., P. Jonsson, S. Pacariz, G. Björk, M. I. Taylor, and H. Svedäng. "Spawning of Kattegat cod (Gadus morhua)—mapping spatial distribution by egg surveys." Fisheries Research 147 (2013): 63-71.

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21.7 Tables and figures

Table 21.1. Whiting in Division 3.a. Landings (t) as supplied by the Study Group on Division 3.a Demersal Stocks (ICES, 1992) and updated by the WGNSSK in 2007. The estimates of discards for 2002–2018 were updated in WKDEM2020 (ICES, 2020).

Year	Denmark*			Norway	Sweden	Others	Total	WG estimate of Discards
1975	19018			57	611	4	19690	
1976	17870			48	1002	48	18968	
1977	18116			46	975	41	19178	
1978	48102			58	899	32	49091	
1979	16971			63	1033	16	18083	
1980	21070			65	1516	3	22654	
	Total consumption	**Total industrial**	**Total**					
1981	1027	23915	24942	70	1054	7	26073	
1982	1183	39758	40941	40	670	13	41664	
1983	1311	23505	24816	48	1061	8	25933	
1984	1036	12102	13138	51	1168	60	14417	

1985	557	11967	12524	45	654	2	13225	
1986	484	11979	12463	64	477	1	13005	
1987	443	15880	16323	29	262	43	16657	
1988	391	10872	11263	42	435	24	11764	
1989	917	11662	12579	29	675	-	13283	
1990	1016	17829	18845	49	456	73	19423	
1991	871	12463	13334	56	527	97	14041	
1992	555	3340	3895	66	959	1	4921	
1993	261	1987	2248	42	756	1	3047	
1994	174	1900	2074	21	440	1	2536	
1995	85	2549	2634	24	431	1	3090	
1996	55	1235	1290	21	182	-	1493	
1997	38	264	302	18	94	-	414	
1998	35	354	389	16	81	-	486	
1999	37	695	732	15	111	-	858	
2000	59	777	836	17	138	1	992	
2001	61	970	1031	27	126	+	1184	
2002	164	1347	1510	23	134	1	1669	2373
2003	104	641	745	20	72	2	839	1837
2004	252	954	1206	17	74	1	1298	2782
2005	110	853	962	13	73	0	1048	1625
2006	71	410	481	11	86	0	578	1497
2007	57	275	332	14	82	1	429	1524
2008	54	286	340	14	52	0	407	795
2009	73	172	245	10	34	0	289	778
2010	49	158	207	10	30	1	248	803
2011	40	44	85	8	20	0	114	937
2012	30	7	37	16	10	1	63	377
2013	29	130	159	8	15	1	183	687
2014	49	346	395	5	37	2	439	649
2015	75	570	645	6	56	5	712	820

2016	129	334	463	13	62	5	543	1307
2017	189	193	382	8	33	7	431	1185
2018	175	156	332	5	34	2	372	1357
2019	78	75	153	5	20	1	179	627
2020	64	437	501	4	14	< 0.5	520	1310
2021	100	85	185	2	12	1	199	1014
2022	193	53	246	3	26	1	276	433
2023	261	44	304	5	27	2	338	388
2024	262	58	319	10	21	2	352	502

* Values from 1992 updated by WGNSSK (2007), WGNSSK (2011).

Table 21.2. Whiting in Division 3.a. Landings and discards split by raised and imported. All weights are in tonnes.

2023	Catch category	Imported or Raised	Catch (tonnes)	Percent
	Landings	Imported	338	
	Discards	Imported	364	94%
	Discards	Raised	24	6%
	Logbook registered discard	Imported	0	
	BMS landing	Imported	0	
2024	**Catch category**	**Imported or Raised**	**Catch (tonnes)**	**Percent**
	Landings	Imported	352	
	Discards	Imported	472	94%
	Discards	Raised	30	6%
	Logbook registered discard	Imported	0	
	BMS landing	Imported	0	

Table 21.3. Whiting in Division 3.a. The predicted biomass index (I) in Quarter 1 between 1983 and 2024 All values are updated each time the biomass index is calculated with new years of data. High and Low refer to 95% confidence intervals.

Year	Biomass index		
	Low	I	High
1983	0.70	0.86	1.05
1984	0.70	0.86	1.06
1985	0.69	0.88	1.12

Year	Biomass index		
	Low	I	High
1986	0.74	0.93	1.16
1987	0.81	1.02	1.28
1988	0.92	1.17	1.48
1989	1.08	1.36	1.72
1990	1.23	1.55	1.95
1991	1.32	1.66	2.09
1992	1.15	1.42	1.76
1993	0.87	1.08	1.33
1994	0.82	1.04	1.33
1995	0.95	1.20	1.51
1996	0.86	1.08	1.37
1997	0.62	0.77	0.95
1998	0.49	0.63	0.82
1999	0.64	0.80	1.00
2000	1.04	1.32	1.69
2001	1.40	1.77	2.25
2002	1.26	1.62	2.07
2003	1.13	1.44	1.83
2004	1.12	1.40	1.77
2005	0.99	1.22	1.51
2006	0.80	1.00	1.24
2007	0.71	0.88	1.09
2008	0.63	0.78	0.96
2009	0.51	0.64	0.80
2010	0.50	0.64	0.82
2011	0.49	0.61	0.76
2012	0.38	0.48	0.60
2013	0.37	0.49	0.64

Year	Biomass index		
	Low	I	High
2014	0.55	0.71	0.91
2015	0.72	0.90	1.12
2016	0.70	0.90	1.14
2017	0.76	0.95	1.20
2018	0.72	0.89	1.10
2019	0.67	0.84	1.04
2020	0.62	0.82	1.09
2021	0.61	0.78	1.01
2022	0.61	0.78	0.99
2023	0.69	0.87	1.10
2024	0.78	0.96	1.16

Table 21.4. Whiting in Division 3.a. Definitions and values of the components of the *rb*-rule.

Component	Definition	Description	Value
A_y	Advice for 2024	Previous catch advice (using the “rb-rule”)	650 tonnes
I_{loss}		Minimum observed index	0.48
$I_{trigger}$	1.4 I_{loss}	Biomass safeguard trigger	0.68
I_{2023}		Last index value	0.69
r	$\frac{\bar{\mathrm{I}}(2022,2023)}{\bar{\mathrm{I}}(2019,2020,2021)}$	Biomass index trend	0.92
b	$\min\left(1, \frac{I_{2023}}{I_{trigger}}\right)$	Biomass safeguard	1
m		Multiplier that ensures probability of B falling below B_{lim} to no more than 5%	0.5
Rule	A_y * r * b * m	“rb-rule”	299 tonnes
Stability clause		+20%/−30% compared to A_y (considered if b=1)	Applied (−30%)
Catch advice for 2025 and 2026			455 tonnes

Table 21.5. Whiting in Division 3.a. ICES advice, agreed total allowable catch (TAC) and ICES estimated landings and discards. All weights are in tonnes.

Year	ICES advice/single-stock exploitation boundaries	Catch corresponding to advice	Landings corresponding to advice	Agreed TAC	ICES landings*	ICES catches*^
1987	Precautionary TAC	-		17000	16657	
1988	Precautionary TAC	-		17000	11764	
1989	Precautionary TAC	-		17000	13283	
1990	Precautionary TAC	-		17000	19423	
1991	TAC	-		17000	14041	
1992	No advice	-		17000	4921	
1993	Precautionary TAC	-		17000	3047	
1994	If required, precautionary TAC	-		17000	2536	
1995	If required, precautionary TAC	-		15200	3090	
1996	If required, precautionary TAC	-		15200	1493	
1997	If required, TAC equal to recent catches	-		15200	414	
1998	No advice			15200	486	
1999	TAC, average period 1993–1996	6000		8000	858	
2000	TAC, average period 1996–1998	1500		4000	992	
2001	TAC, average period 1996–1998	1500		2500	1184	
2002	TAC, average period 1996–1998	1500		2000	1669	4042

Year	ICES advice/single-stock exploitation boundaries	Catch corresponding to advice	Landings corresponding to advice	Agreed TAC	ICES landings*	ICES catches*^
2003	TAC, average period 1996–1998	1500		1500	839	2676
2004	TAC, average period 1996–1998	1500		1500	1298	4080
2005	Average period 1996–1998	1500		1500	1048	2673
2006	Average period 1996–1998	1500		1500	578	2075
2007	Average period 1996–1998	1500		1500	429	1953
2008	Recent average catches	1050		1050	407	1201
2009	Same advice as last year	1050		1050	289	1067
2010	Same advice as last year	1050		1050	248	1050
2011	No advice	-		1050	114	1050
2012	Reduce catch	-		1050	63	440
2013	20% Reduction in catches (average of the last 3 years)	< 500		1050	183	870
2014	No new advice, same as for 2013	< 500		1050	439	1088
2015	No new advice, same as for 2014	< 500	< 212	1050	712	1531
2016	Precautionary approach (same advised catch value as given for 2015)	≤ 500		1050	543	1850
2017	Precautionary approach (same advised catch value as given for 2015)	≤ 500		1050	431	1615
2018	Precautionary approach	≤ 400		1050	372	1729
2019	Precautionary approach	≤ 400		1660	179	806

Year	ICES advice/single-stock exploitation boundaries	Catch corresponding to advice	Landings corresponding to advice	Agreed TAC	ICES landings*	ICES catches*^
2020	Precautionary approach	≤ 400		1660	520	1830
2021	Precautionary approach	≤ 929		929	199	1212
2022	Precautionary approach	≤ 929		929	276	709
2023	Precautionary approach	≤ 676		676	338	726
2024	Precautionary approach	≤ 650^^		676	352	854
2025	Precautionary approach	≤ 455		455		
2026	Precautionary approach	≤ 455				

*** Includes bycatch in small-mesh industrial fishery.**

^ Since 2018, discards include BMS landings

^^ The advice of ≤ 676 tonnes for 2024 (originally published in 2022) was updated in 2024.

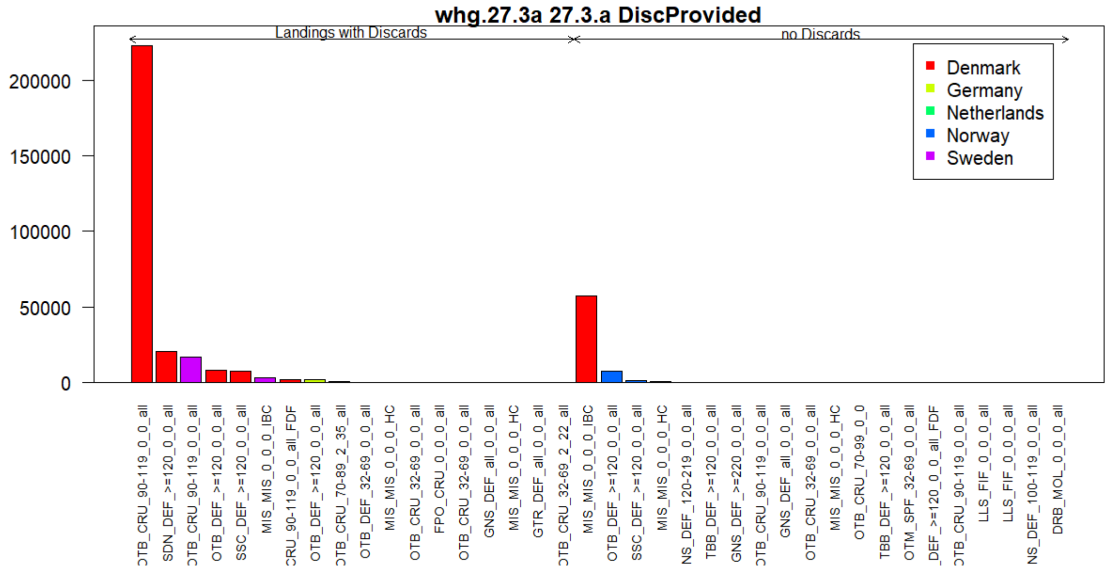

Figure 21.1. Whiting in Division 3.a. Summary of the information provided to InterCatch for 2024. Landings by metier and country, distinguishing between strata with and without corresponding discard information provided.

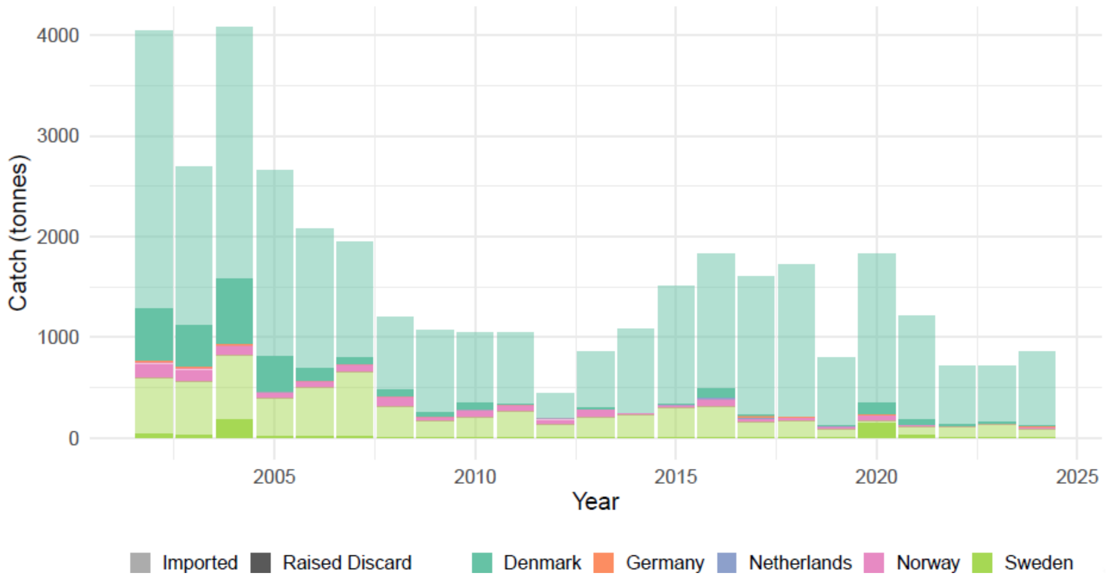

Figure 21.2. Whiting in Division 3.a. Summary of the information provided to InterCatch since 2002. Catch by country, distinguishing between imported catch (landings + discards) and raised catch (discards only).

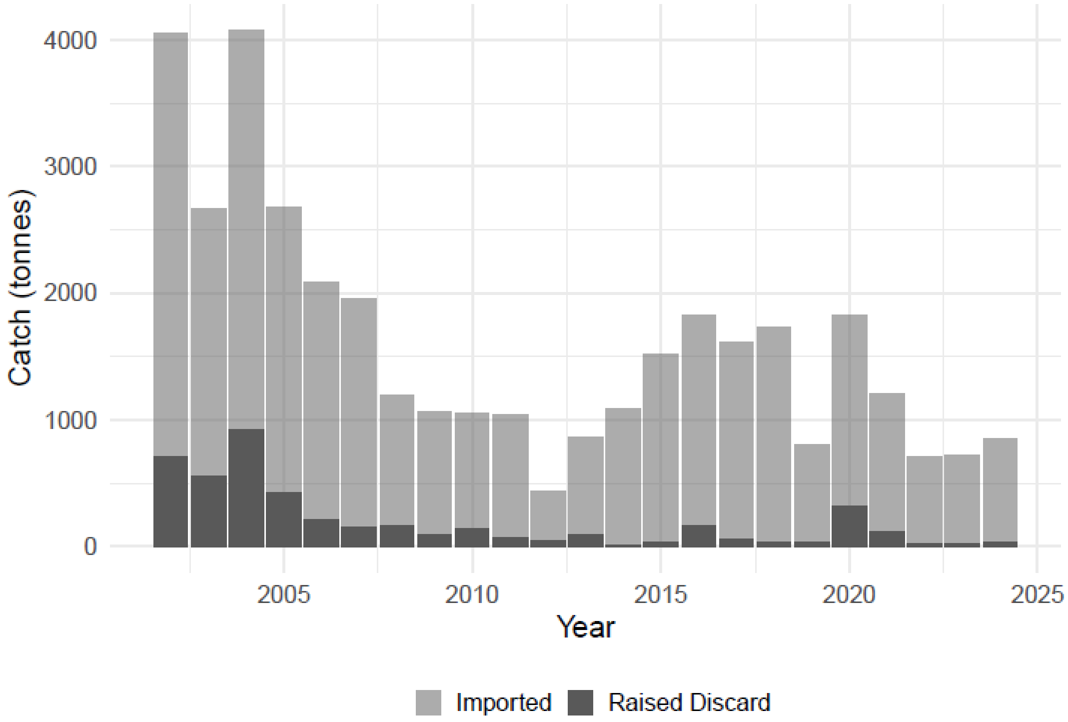

Figure 21.3. Whiting in Division 3.a. Summary of the catch provided to InterCatch since 2002 distinguishing between imported catch (landings + discards) and raised catch (discards only).

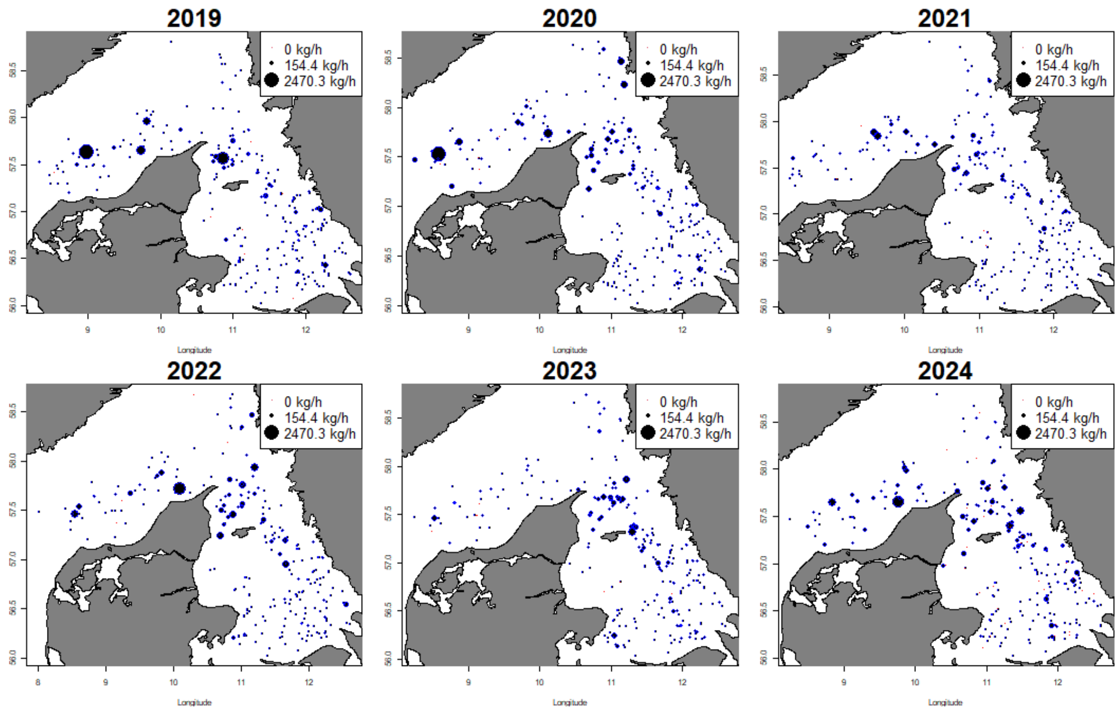

Figure 21.4. Whiting in Division 3.a. Observed whiting biomass in kg per hour between 2019 and 2024 from the four bottom trawl surveys used to estimate a biomass index (North Sea International Bottom Trawl Survey, the Baltic International Trawl Survey, and two Danish national surveys that target sole and cod). Each dot is a survey haul and the size of the dot scales with biomass (scales vary between years).

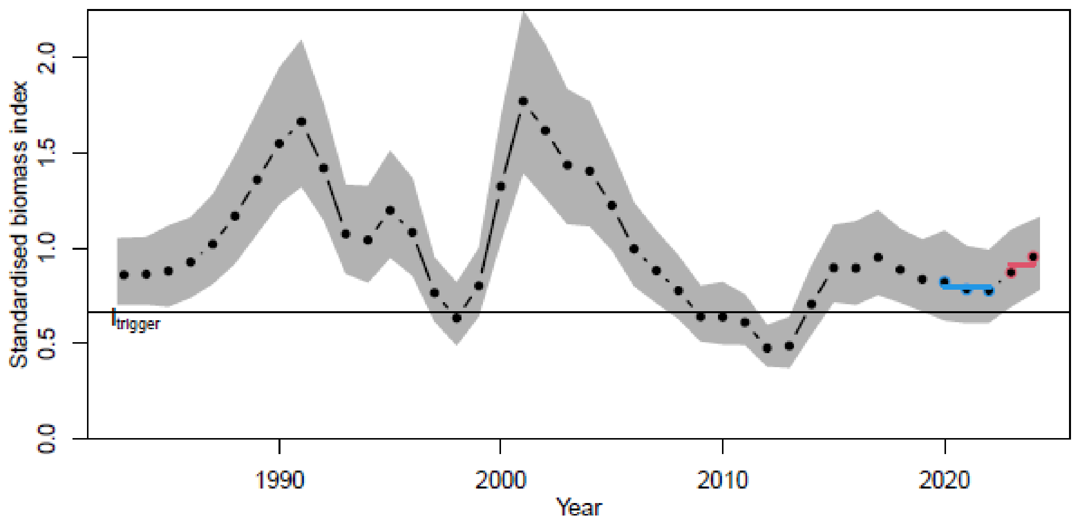

Figure 21.5. Whiting in Division 3.a. Combined biomass index (Q1) using survey data from the two international bottom trawl surveys and two Danish national surveys. The average of the last two years (red line) and the average of the three years before that (blue line) are used to calculate the *r* multiplier of the "*rb-rule*". The horizontal line shows the $I_{trigger}$ reference point which is 1.4 times the minimum observed index value (I_{loss}). The shaded grey area shows the 95% confidence interval.

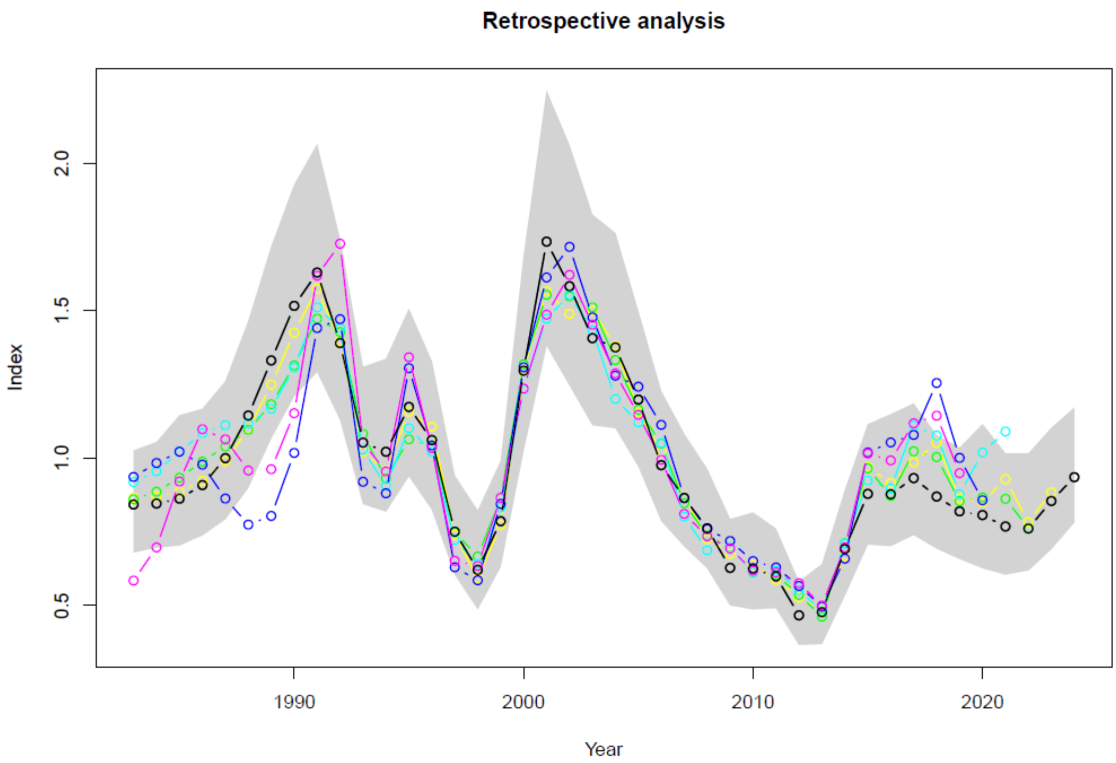

Figure 21.6. Whiting in Division 3.a. Retrospective analysis of the standardised quarter 1 biomass index.

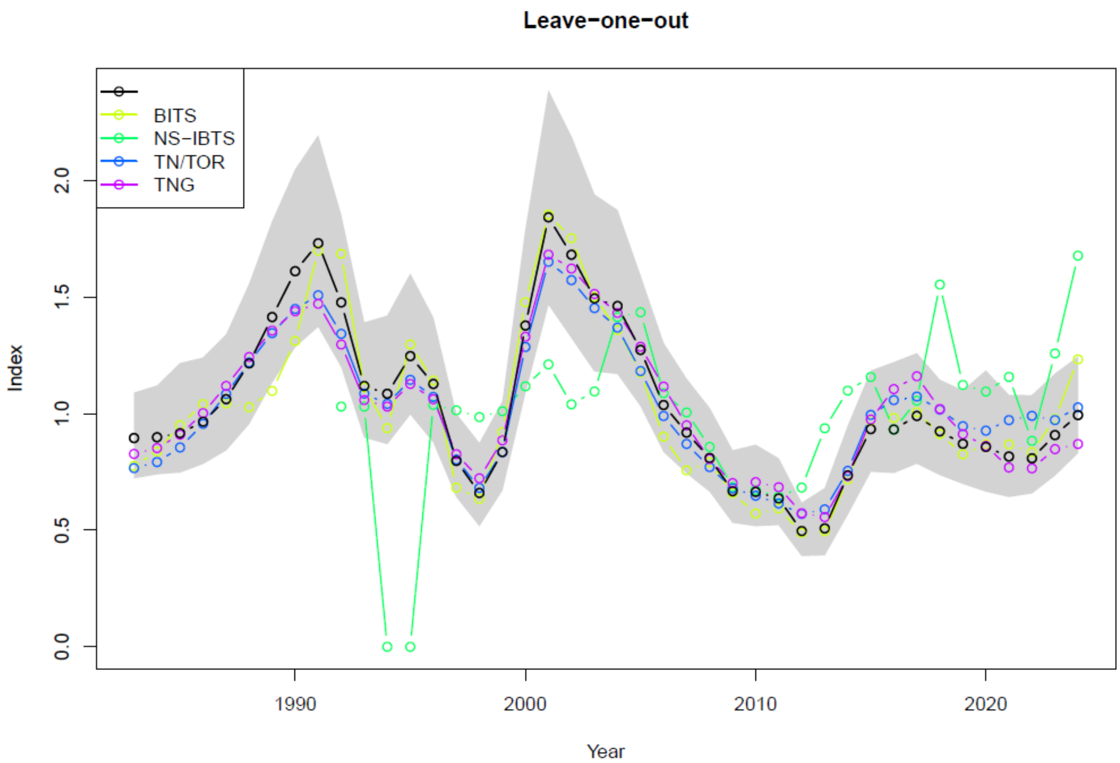

Figure 21.7. Whiting in Division 3.a. Leave-one-survey-out analysis of the standardised quarter 1 biomass index.

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